

PRATIK RATHORE

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Education

Stanford University

PhD Candidate in Electrical Engineering

September 2021 – June 2026 (expected)

Stanford, CA

University of Maryland

B.S. in Electrical Engineering (summa cum laude), B.S. in Mathematics (summa cum laude)

August 2017 – May 2021

College Park, MD

Experience

Stanford University, Department of Management Science & Engineering

June 2022 – Present

Research Assistant (advised by Madeleine Udell)

Stanford, CA

- Developing optimization algorithms leveraging randomized numerical linear algebra to address scalability and stability challenges in training machine learning models
- Creating and maintaining high-quality, open-source implementations of these methods; adopted in popular machine learning libraries such as DeepXDE
- Applying scientific machine learning to solve PDE-governed problems in geophysics

Skyworks Solutions, Technology & Manufacturing Data Analytics Team

June 2025 – August 2025

Machine Learning & AI Intern

Irvine, CA

- Led development of a Python library that automates circuit topology generation and simulation configuration for designing radio frequency (RF) filters, reducing design times from one to two weeks to 7-8 hours
- Collaborated with software engineers to build a web application allowing circuit designers to interface with the automated topology library
- Designed an AI-driven circuit design automation system and implemented prototype workflows

Gridmatic, Power Trading & Optimization Team

June 2024 – September 2024

Research Scientist Intern

Cupertino, CA

- Applied scenario reduction to speed up linear programs in battery scheduling, while preserving profits
- Developed a new backtest framework that accounts for Gridmatic's price impact in ERCOT market
- Formulated, implemented, and tested price impact models based on residual demand curves
- Proposed an ADMM-based algorithm for price impact-aware portfolio optimization

Selected Publications

J. Stitt, **P. Rathore**, M. Udell, and C.-Y. Lai. *SubsurfaceGen: Procedural Generation of Field-Scale Earth Models and Seismic Data*. Submitted, 2026, arxiv:2605.30541

P. Rathore, Z. Frangella, S. Garg, S. Fazliani, M. Dereziński, and M. Udell. *Turbocharging Gaussian Process Inference with Approximate Sketch-and-Project*. NeurIPS, 2025, arxiv:2505.13723

P. Rathore, Z. Frangella, J. Yang, M. Dereziński, and M. Udell. *Have ASkotch: A Neat Solution for Large-scale Kernel Ridge Regression*. Submitted, 2025, arxiv:2407.10070

P. Rathore, W. Lei, Z. Frangella, L. Lu, and M. Udell. *Challenges in Training PINNs: A Loss Landscape Perspective*. ICML, 2024, arxiv:2402.01868 (**Oral, top 1.5% of all submissions**)

Z. Frangella*, **P. Rathore***, S. Zhao, and M. Udell.¹ *PROMISE: Preconditioned Stochastic Optimization Methods by Incorporating Scalable Curvature Estimates*. JMLR, 2024, arxiv:2309.02014

Awards & Honors

University of Maryland Department of Mathematics High Honors Medal

May 2021

NSF Graduate Research Fellowship Honorable Mention

March 2021

University of Maryland Department of Electrical and Computer Engineering Chair's Award

March 2021

University of Maryland Dan Shanks Award for research in number theory

April 2019

University of Maryland Putnam Team Member (individually ranked top 3% in 2019, top 5% in 2020)

2019, 2020

Banneker-Key Scholar – a full merit scholarship awarded to top 1% of undergraduates

August 2017 – May 2021

United States of America Mathematical Olympiad (USAMO) Qualifier

May 2017

Programming Skills

Proficient: Python, PyTorch, NumPy, MATLAB, $\text{\LaTeX}$

Familiar: Pandas, C/C++, Julia, Java, R, Simulink

Relevant Coursework

Machine Learning, Machine Learning for Sequence Modeling, Machine Learning for Discrete Optimization, Reinforcement Learning, Convex Optimization, Theory of Statistics, Numerical Linear Algebra, Parallel Computing

^{1*} denotes equal contribution.